

Sekar M

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RESEARCH INTERESTS

Natural Language Processing, Large Language Models, Reasoning for Language Models, Indian Legal NLP & Question Answering, Mixture-of-Experts (MoE), Multilingual and Multi-modal AI

TECHNICAL SKILLS

- **Languages & Databases:** Python, C, C++, Java, Neo4j, Hadoop
- **Frameworks & Libraries:** PyTorch, HuggingFace (Transformers/PEFT), LangGraph, scikit-learn, NumPy, NLTK
- **Deep Learning & LLMs:** Fine-tuning (LoRA/DoRA), Mixture-of-Experts (MoE), Graph Neural Networks (GNNs)
- **Tools & Systems:** Linux, Git, LaTeX, Docker

EDUCATION

National Institute of Technology Tiruchirappalli, Tamil Nadu, India Ph.D. in Computer Science and Engineering	July 2024 – Present CGPA: 8.5/10
Government College of Engineering, Bargur (Anna University), Tamil Nadu, India M.E. in Computer Science and Engineering	July 2022 – May 2024 CGPA: 8.0/10
Varuvan Vadivelan Institute of Technology (Anna University), Tamil Nadu, India B.E. in Computer Science and Engineering	August 2018 – May 2022 CGPA: 7.5/10

PUBLICATIONS

- Sekar M, Santhanavijayan A. *H-MoLE: Hierarchical Mixture of Legal Experts for Indian Legal Reasoning and Question Answering*. Workshop on Natural Legal Language Processing (NLLP @ EMNLP 2026). (Under Review)
 - Sekar M, Pranava Mohan, Santhanavijayan A. *When Models Defer: Authority Framing Drives Social Compliance, Not Re-Evaluation, in LLM Multi-Agent Critique*. Submitted to EMNLP 2026 (via ACL ARR May 2026). (Under Review)
 - Sekar M, Jatavath Sirisha, Santhanavijayan A. *TalkingToM: Dual-Process Cognitive Modeling for Narrative Theory of Mind Question Answering*. 7th International Conference on Speech and Language Technologies for Low-Resource Languages (SPELLL 2026). (Under Review)
 - Pranava Mohan, Sekar M, Santhanavijayan A. *tensorveeras@CC-MMD 2026: Cross-Cultural Misogynistic Meme Detection using a Multimodal Mixture of Experts*. ACM ICMI 2026 Grand Challenge: CC-MMD 2026. (Accepted)
 - Sekar M, Sirisha Jatavath, Santhanavijayan A. *ViBrainNet: An Attention-Based CNN-Vision Transformer Ensemble for Brain Tumor Detection and Segmentation in MRI*. 2nd Congress on Intelligent Machines and Algorithms (CIMA 2026). (Accepted and Presented)
 - Sekar M, J. Nafeesa Begum, K. Kumar, S. Selvi. *RowColumnPrune: An Efficient Compression of Large Language Models through Row and Column Deletion*. 10th International Conference on Latest Trends in Science, Engineering and Technology (ICLTSET'24), April 2024. ISBN 978-93-6228-897-4.
 - Sekar M, Chandrakanth K, Sivasakthi R, Vasanthakumar M. *A Machine Learning Approach to Dimensionality Reduction in Large Data sets*. National Conference on Computing & Information System (NCCIS'22), May 2022.
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ACADEMIC EXPERIENCE

Department of CSE, NIT Tiruchirappalli
Ph.D. Research and Teaching Experience (HTRA)

Tiruchirappalli, Tamil Nadu, India
Jul'24 – Present

- **Course Instructor, CSHO12 – Distributed Systems** Jun'25 – Dec'25
- **Course Instructor, CSIR15 – Essentials of Computer Science** Jan'26 – May'26
- **Laboratory Teaching Assistant, CSLR62 – Web Technology Laboratory**

Capgemini
Software Developer Intern

Bengaluru, Karnataka, India
Feb'22 – Jun'22

CONFERENCES & WORKSHOPS

6th Indian Symposium on Machine Learning (IndoML 2025)

BITS Pilani Hyderabad, India

Graduate Forum – Poster Presentation: *TamilaAR: A Tamil Analogy Test Set for Multilingual Analogical Reasoning in Large Language Models*
Accepted

21st International Conference on Natural Language Processing (ICON 2024)

Chennai, India

NLP Association of India (NLP AI) – AU-KBC Research Centre

19th – 22nd December 2024

Art of Computing Workshop

Bengaluru, India

Department of CSA, Indian Institute of Science

17th – 18th February 2024

Lecture: How Could Machines Reach Human-Level Intelligence?

Chennai, India

Speaker: Yann LeCun (VP & Chief AI Scientist, Meta) – Indian Institute of Technology Madras

22nd October 2024

ACM Annual Event

Coimbatore, India

PSG College of Technology

1st March 2025

ACHIEVEMENT

- **Overall 2nd Rank in the Cross-Cultural Misogynistic Meme Detection (CC-MMD 2026) Grand Challenge**, co-located with the ACM International Conference on Multimodal Interaction (ICMI 2026).